INTRODUCTION TO INFORMATION GEOMETRY

# Fisher Information and Scores

## Week 7 · Lecture 1

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# A metric from the KL expansion

For positive $P = (p_i)$ and a small tangent displacement $u$,

$$D(P\|P+u) = \frac{1}{2}\sum_i \frac{u_i^2}{p_i} + O(\|u\|^3).$$

The resulting Fisher inner product is

$$g_P(u,v) = \sum_i \frac{u_i v_i}{p_i}, \qquad \sum_i u_i = \sum_i v_i = 0.$$

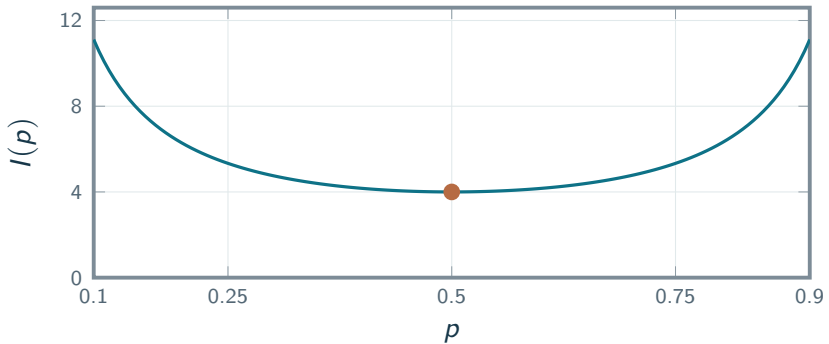
# Length depends on the base distribution

For the Bernoulli family, a probability displacement $dp$ has squared length

$$ds^2 = \frac{dp^2}{p(1-p)}.$$

The same change in $p$ carries more Fisher length near a boundary than near $p = \frac{1}{2}$.

# The Bernoulli metric coefficient




Here $I(p) = 1/[p(1-p)]$; it is smallest at $p = 1/2$.

# A Fisher norm on the triangle

At $P = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, consider $v = (0, .1, -.1)$. Find the squared Fisher norm

$$g_P(v, v) = \sum_i \frac{v_i^2}{p_i} = ?.$$

# A Fisher norm on the triangle

At $P = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, consider $v = (0, .1, -.1)$. The squared norm is

$$g_P(v, v) = \frac{.01}{.25} + \frac{.01}{.25} = .08.$$

# Scores describe parameter sensitivity

Let $P_\theta = (p_i(\theta))$ be a statistical family. For observed outcome $i$, the score is

$$S_\theta(i) = \frac{\partial}{\partial\theta} \ln p_i(\theta).$$

Its mean is zero:

$$E_\theta[S_\theta] = \sum_i p_i \frac{p_i'}{p_i} = 0.$$

# Fisher information is score variance

For a one-parameter family,

$$I(\theta) = E_\theta[S_\theta^2] = \sum_i \frac{(p_i'(\theta))^2}{p_i(\theta)}.$$

This is precisely the squared Fisher speed of the parameter curve. For several parameters, the expected score products form the matrix $G_{ab} = E[S_a S_b]$.

# Information in one Bernoulli observation

For outcomes $0, 1$, let $P_p = (1-p, p)$. Find the two scores and then compute

$$I(p) = E_p[S_p^2].$$

$$S_p(1) = ?, \quad S_p(0) = ?, \quad I(p) = ?.$$

# Information in one Bernoulli observation

For outcomes $0, 1$, let $P_p = (1-p, p)$. The scores are $\frac{1}{p}$ for outcome 1 and $-\frac{1}{1-p}$ for outcome 0.

$$I(p) = p\left(\frac{1}{p}\right)^2 + (1-p)\left(\frac{1}{1-p}\right)^2 = \frac{1}{p(1-p)}.$$

# Information under a coordinate change

Suppose $\theta = \theta(\eta)$. The score obeys the chain rule. For a one-dimensional family,

$$I_\eta = I_\theta \left( \frac{d\theta}{d\eta} \right)^2 .$$

In several dimensions, with Jacobian $J$,

$$G_\eta = J^T G_\theta J.$$

# Information in log-odds coordinates

Let $L = \ln[p/(1-p)]$. Using $dp/dL = p(1-p)$ and $I_p = 1/[p(1-p)]$, find $I_L$.

$$I_L = ?.$$

# Information in log-odds coordinates

Let $L = \ln[p/(1-p)]$. The transformation law gives

$$I_L = \frac{1}{p(1-p)}[p(1-p)]^2 = p(1-p).$$

The metric coefficient changes with the coordinate.

# Information accumulates in a sample

For $N$ independent observations, log likelihoods add. Their score contributions have mean zero, so cross terms vanish:

$$I_N(\theta) = NI(\theta).$$

The associated local statistical lengths scale by $\sqrt{N}$.

# Cramér–Rao and statistical precision

Under the usual unbiasedness and differentiability hypotheses, an estimator $T$ of $\theta$ satisfies

$$\operatorname{Var}_\theta(T) \geq \frac{1}{N I(\theta)}.$$

For Bernoulli sampling, the empirical proportion has variance $p(1-p)/N$ and attains this bound.

# A lower bound for an estimator

Take $N = 100$ independent Bernoulli observations at $p = 0.2$. Compute $I(p)$ and the Cramér–Rao lower bound on variance:

$$I(p) = ?, \qquad \mathrm{Var}(T) \geq ?.$$

# A lower bound for an estimator

Take $N = 100$ independent Bernoulli observations at $p = 0.2$.

$$I(p) = \frac{1}{.2(.8)} = 6.25.$$

$$\mathrm{Var}(T) \geq \frac{1}{625} = .0016.$$

# The same object in three languages

- The Hessian of local KL divergence defines a quadratic form.
- Tangent vectors have lengths measured by that form.
- A statistical score has expected square equal to the same length.